Richness plots

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```r
# libraries ----
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)

# data ----

data <- map(1:5, \(x) {
  #read all three taxa
  read_excel("Data2024.xlsx", sheet = x) |>
    #pivot
    pivot_longer(-1, names_to = "species") |>
    #rename first column
    rename(plot = 1)
    }
  )

#first three sheets are vascular plants, join them together
data <- bind_rows(data[1:3]) |>
  list() |>
  c(data[4:5])

data <- data |>
  #bind list to single df
  bind_rows(.id = "taxon") |>
  #rename IDs
  mutate(taxon = fct_recode(taxon,
                            "Vascular plants" = "1",
                            "Saproxylic fungi" = "2",
                            "Epiphytic lichens" = "3") |>
```

```r
          fct_relevel("Saproxylic fungi", after = 2),
        #extract genus
        genus = word(species, 1),
        #extract epithets
        epit = word(species, 2),
        #extract site
        site = substr(plot, 1, 2) |>
          fct_recode("Prati di Tivo" = "PR",
                     "Incodaro" = "IN",
                     "Venaquaro" = "VE")
  ) |>
  arrange(taxon, species) |>
  #these were wide matrices, keep only observed entries
  filter(value > 0)

#distinct species names
distinct(data, species) |> pull() |> sort()
```

```
##   [1] "Abies alba"                 "Acer campestre"
##   [3] "Acer opalus"                "Acer platanoides"
##   [5] "Acer pseudoplatanus"        "Actaea spicata"
##   [7] "Adenostyles australis"      "Adoxa moschatellina"
##   [9] "Aegopodium podagraria"      "Agrimonia eupatoria"
##  [11] "Agrostis capillaris"        "Ajuga reptans"
##  [13] "Allium ursinum"             "Anaptychia ciliaris"
##  [15] "Anaptychia crinalis"        "Aquilegia dumeticola"
##  [17] "Arabis alpina"              "Arabis collina"
##  [19] "Arabis hirsuta"             "Arctium lappa"
##  [21] "Arctium minus"              "Aremonia agrimonoides"
##  [23] "Armillaria sp."             "Arthonia didyma"
##  [25] "Arthonia radiata"           "Arum maculatum"
##  [27] "Asperula laevigata"         "Asplenium ruta-muraria"
##  [29] "Asplenium trichomanes"      "Astragalus glycyphyllos"
##  [31] "Athallia cerinelloides"     "Athallia pyracea"
##  [33] "Atropa bella-donna"         "Avenella flexuosa"
##  [35] "Bertia moriformis"          "Biscogniauxia nummularia"
##  [37] "Bjerkandera adusta"         "Brachypodium rupestre"
##  [39] "Brachypodium sylvaticum"    "Bromopsis ramosa"
##  [41] "Buellia griseovirens"       "Bupleurum falcatum"
##  [43] "Caloplaca cerina"           "Caloplaca sp."
##  [45] "Calycina citrina"           "Campanula glomerata"
##  [47] "Campanula persicifolia"     "Campanula rapunculus"
##  [49] "Campanula tanfanii"         "Campanula trachelium"
##  [51] "Candelariella faginea"      "Cardamine bulbifera"
##  [53] "Cardamine enneaphyllos"     "Cardamine graeca"
##  [55] "Cardamine heptaphylla"      "Cardamine kitaibelii"
##  [57] "Carduus sp."                "Carex digitata"
##  [59] "Carex divulsa"              "Carex flacca"
##  [61] "Carex pilosa"               "Carex remota"
##  [63] "Carex sylvatica"            "Centaurium erythraea"
##  [65] "Cephalanthera damasonium"   "Cephalanthera longifolia"
##  [67] "Cephalanthera rubra"        "Cerioporus varius"
##  [69] "Chlorociboria aeruginascens" "Chondrostereum purpureum"
```

```
##  [71] "Circaea lutetiana"           "Cirsium arvense"
##  [73] "Clematis vitalba"            "Clinopodium menthifolium"
##  [75] "Clinopodium nepeta"          "Clinopodium vulgare"
##  [77] "Coniophora sp."              "Crataegus monogyna"
##  [79] "Cruciata glabra"             "Cruciata laevipes"
##  [81] "Crustoderma sp."             "Cyclamen repandum"
##  [83] "Cymbalaria muralis"          "Cynosurus cristatus"
##  [85] "Cystopteris fragilis"        "Cytisophyllum sessilifolium"
##  [87] "Cytisus scoparius"           "Cytisus villosus"
##  [89] "Dactylis glomerata"          "Dactylorhiza maculata"
##  [91] "Daphne laureola"             "Daphne mezereum"
##  [93] "Dianthus sylvestris"         "Diatrype disciformis"
##  [95] "Diatrype stigma"             "Digitalis micrantha"
##  [97] "Doronicum columnae"          "Drymochloa sylvatica"
##  [99] "Dryopteris filix-mas"        "Epilobium montanum"
## [101] "Epipactis helleborine"       "Epipactis microphylla"
## [103] "Epitrachys italica"          "Euonymus europaeus"
## [105] "Euphorbia amygdaloides"      "Euphorbia cyparissias"
## [107] "Euphrasia stricta"           "Fagus sylvatica"
## [109] "Festuca heterophylla"        "Fomes fomentarius"
## [111] "Fomitopsis pinicola"         "Fragaria vesca"
## [113] "Fraxinus excelsior"          "Galerina marginata"
## [115] "Galium aparine"              "Galium lucidum"
## [117] "Galium odoratum"             "Galium rotundifolium"
## [119] "Ganoderma applanatum"        "Geranium nodosum"
## [121] "Geranium purpureum"          "Geranium robertianum"
## [123] "Glaucomaria carpinea"        "Grafia golaka"
## [125] "Gyalecta sp."                "Gyalolechia flavorubescens"
## [127] "Hedera helix"                "Helianthemum nummularium"
## [129] "Helleborus viridis"          "Helminthotheca echioides"
## [131] "Hepatica nobilis"            "Heterodermia speciosa"
## [133] "Hieracium murorum"           "Hordelymus europaeus"
## [135] "Hypericum montanum"          "Hypericum perforatum"
## [137] "Hypoxylon fragiforme"        "Hypoxylon sp."
## [139] "Ilex aquifolium"             "Juniperus communis"
## [141] "Laburnum anagyroides"        "Lachnum relicinum"
## [143] "Lapsana communis"            "Laserpitium latifolium"
## [145] "Lathyrus pratensis"          "Lathyrus vernus"
## [147] "Lecania naegelii"            "Lecanora albella"
## [149] "Lecanora argentata"          "Lecanora chlarotera"
## [151] "Lecanora intumescens"        "Lecidella elaeochroma"
## [153] "Lenzites betulinus"          "Lepra amara"
## [155] "Lepraria rigidula"           "Lepraria sp."
## [157] "Lilium bulbiferum"           "Lilium martagon"
## [159] "Limodorum abortivum"         "Linum viscosum"
## [161] "Lonicera alpigena"           "Lonicera etrusca"
## [163] "Lotus corniculatus"          "Luzula forsteri"
## [165] "Luzula multiflora"           "Luzula sylvatica"
## [167] "Medicago lupulina"           "Megacollybia platyphylla"
## [169] "Melanelixia glabratula"      "Melica uniflora"
## [171] "Milium effusum"              "Moehringia trinervia"
## [173] "Monotropa hypophegea"        "Mycelis muralis"
## [175] "Mycena renati"               "Nemania serpens"
## [177] "Neottia nidus-avis"          "Ochrolechia pallescens"
```

```
## [179] "Opegrapha sp."                 "Orthilia secunda"
## [181] "Ostrya carpinifolia"           "Oxalis acetosella"
## [183] "Parmelia sulcata"              "Parmelina spp."
## [185] "Pertusaria coccodes"           "Pertusaria flavida"
## [187] "Pertusaria leioplaca"          "Pertusaria pertusa"
## [189] "Petrosedum rupestre"           "Phaeophyscia insignis"
## [191] "Phlyctis agelaea"              "Phlyctis argena"
## [193] "Physcia adscendens"            "Physcia aipolia"
## [195] "Physcia biziana"               "Physcia leptalea"
## [197] "Physcia sp."                   "Physcia stellaris"
## [199] "Physconia distorta"            "Picris hieracioides"
## [201] "Platanthera bifolia"           "Pleurosticta acetabulum"
## [203] "Pleurotus cornucopiae"         "Pluteus cervinus"
## [205] "Poa compressa"                 "Poa nemoralis"
## [207] "Poa trivialis"                 "Polygonatum odoratum"
## [209] "Polyozosia hagenii"            "Polyporus tuberaster"
## [211] "Polystichum aculeatum"         "Polystichum setiferum"
## [213] "Potentilla micrantha"          "Prenanthes purpurea"
## [215] "Primula vulgaris"              "Prunella vulgaris"
## [217] "Prunus avium"                  "Prunus sp."
## [219] "Pteridium aquilinum"           "Ptilostemon strictus"
## [221] "Pulmonaria hirta"              "Ramalina farinacea"
## [223] "Ramalina fraxinea"             "Ranunculus lanuginosus"
## [225] "Rhamnus alpina"                "Rosa arvensis"
## [227] "Rosa pendulina"                "Rubus hirtus"
## [229] "Rubus idaeus"                  "Rumex acetosa"
## [231] "Ruscus hypoglossum"            "Salix caprea"
## [233] "Sambucus ebulus"               "Sanicula europaea"
## [235] "Saxifraga paniculata"          "Saxifraga rotundifolia"
## [237] "Schizopora paradoxa"           "Scleroderma verrucosum"
## [239] "Scoliciosporum chlorococcum" "Scutellinia sp."
## [241] "Sedum cepaea"                  "Senecio nemorensis"
## [243] "Sesleria autumnalis"           "Sesleria nitida"
## [245] "Silene italica"                "Solidago virgaurea"
## [247] "Sorbus aria"                   "Sorbus aucuparia"
## [249] "Stachys sylvatica"             "Steccherinum robustius"
## [251] "Stereum hirsutum"              "Stereum sp."
## [253] "Stereum subtomentosum"         "Tanacetum corymbosum"
## [255] "Taxus baccata"                 "Tephromela atra"
## [257] "Thesium sp."                   "Trametes gibbosa"
## [259] "Trametes hirsuta"              "Trametes ochracea"
## [261] "Trametes sp."                  "Trametes versicolor"
## [263] "Trifolium campestre"           "Trifolium ochroleucon"
## [265] "Trifolium pratense"            "Trifolium repens"
## [267] "Urtica dioica"                 "Ustulina deusta"
## [269] "Verbascum sp."                 "Veronica chamaedrys"
## [271] "Veronica cymbalaria"           "Veronica hederifolia"
## [273] "Veronica montana"              "Veronica officinalis"
## [275] "Vicia cracca"                  "Vicia grandiflora"
## [277] "Vicia lathyroides"             "Viola alba"
## [279] "Viola reichenbachiana"         "Xanthoria parietina"
## [281] "Xerocomus subtomentosus"       "Xylaria carpophila"
## [283] "Xylaria hypoxylon"             "Xylaria longipes"
## [285] "Xylaria polymorpha"            "Xylaria sp."
```

```
## [287] "Xylariaceae sp."
```

```r
#taxon species richness
data |>
  group_by(taxon) |>
  summarise(n_distinct(species), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon             `n_distinct(species)`
##   <fct>                             <int>
## 1 Vascular plants                     192
## 2 Epiphytic lichens                    49
## 3 Saproxylic fungi                     46
```

```r
#distinct genera
distinct(data, genus) |> pull() |> sort()
```

```
##   [1] "Abies"           "Acer"            "Actaea"          "Adenostyles"
##   [5] "Adoxa"           "Aegopodium"      "Agrimonia"       "Agrostis"
##   [9] "Ajuga"           "Allium"          "Anaptychia"      "Aquilegia"
##  [13] "Arabis"          "Arctium"         "Aremonia"        "Armillaria"
##  [17] "Arthonia"        "Arum"            "Asperula"        "Asplenium"
##  [21] "Astragalus"      "Athallia"        "Atropa"          "Avenella"
##  [25] "Bertia"          "Biscogniauxia"   "Bjerkandera"     "Brachypodium"
##  [29] "Bromopsis"       "Buellia"         "Bupleurum"       "Caloplaca"
##  [33] "Calycina"        "Campanula"       "Candelariella"   "Cardamine"
##  [37] "Carduus"         "Carex"           "Centaurium"      "Cephalanthera"
##  [41] "Cerioporus"      "Chlorociboria"   "Chondrostereum"  "Circaea"
##  [45] "Cirsium"         "Clematis"        "Clinopodium"     "Coniophora"
##  [49] "Crataegus"       "Cruciata"        "Crustoderma"     "Cyclamen"
##  [53] "Cymbalaria"      "Cynosurus"       "Cystopteris"     "Cytisophyllum"
##  [57] "Cytisus"         "Dactylis"        "Dactylorhiza"    "Daphne"
##  [61] "Dianthus"        "Diatrype"        "Digitalis"       "Doronicum"
##  [65] "Drymochloa"      "Dryopteris"      "Epilobium"       "Epipactis"
##  [69] "Epitrachys"      "Euonymus"        "Euphorbia"       "Euphrasia"
##  [73] "Fagus"           "Festuca"         "Fomes"           "Fomitopsis"
##  [77] "Fragaria"        "Fraxinus"        "Galerina"        "Galium"
##  [81] "Ganoderma"       "Geranium"        "Glaucomaria"     "Grafia"
##  [85] "Gyalecta"        "Gyalolechia"     "Hedera"          "Helianthemum"
##  [89] "Helleborus"      "Helminthotheca"  "Hepatica"        "Heterodermia"
##  [93] "Hieracium"       "Hordelymus"      "Hypericum"       "Hypoxylon"
##  [97] "Ilex"            "Juniperus"       "Laburnum"        "Lachnum"
## [101] "Lapsana"         "Laserpitium"     "Lathyrus"        "Lecania"
## [105] "Lecanora"        "Lecidella"       "Lenzites"        "Lepra"
## [109] "Lepraria"        "Lilium"          "Limodorum"       "Linum"
## [113] "Lonicera"        "Lotus"           "Luzula"          "Medicago"
## [117] "Megacollybia"    "Melanelixia"     "Melica"          "Milium"
## [121] "Moehringia"      "Monotropa"       "Mycelis"         "Mycena"
## [125] "Nemania"         "Neottia"         "Ochrolechia"     "Opegrapha"
## [129] "Orthilia"        "Ostrya"          "Oxalis"          "Parmelia"
## [133] "Parmelina"       "Pertusaria"      "Petrosedum"      "Phaeophyscia"
## [137] "Phlyctis"        "Physcia"         "Physconia"       "Picris"
## [141] "Platanthera"     "Pleurosticta"    "Pleurotus"       "Pluteus"
```

```
## [145] "Poa"            "Polygonatum"    "Polyozosia"     "Polyporus"
## [149] "Polystichum"    "Potentilla"     "Prenanthes"     "Primula"
## [153] "Prunella"       "Prunus"         "Pteridium"      "Ptilostemon"
## [157] "Pulmonaria"     "Ramalina"       "Ranunculus"     "Rhamnus"
## [161] "Rosa"           "Rubus"          "Rumex"          "Ruscus"
## [165] "Salix"          "Sambucus"       "Sanicula"       "Saxifraga"
## [169] "Schizopora"     "Scleroderma"    "Scoliciosporum" "Scutellinia"
## [173] "Sedum"          "Senecio"        "Sesleria"       "Silene"
## [177] "Solidago"       "Sorbus"         "Stachys"        "Steccherinum"
## [181] "Stereum"        "Tanacetum"      "Taxus"          "Tephromela"
## [185] "Thesium"        "Trametes"       "Trifolium"      "Urtica"
## [189] "Ustulina"       "Verbascum"      "Veronica"       "Vicia"
## [193] "Viola"          "Xanthoria"      "Xerocomus"      "Xylaria"
## [197] "Xylariaceae"
```

```r
#taxon genus richness
data |>
  group_by(taxon) |>
  summarise(n_distinct(genus), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon             `n_distinct(genus)`
##   <fct>                           <int>
## 1 Vascular plants                   132
## 2 Epiphytic lichens                  31
## 3 Saproxylic fungi                   34
```

```r
#distinct epithets
distinct(data, epit) |> pull() |> sort()
```

```
##   [1] "abortivum"       "acetabulum"      "acetosa"         "acetosella"
##   [5] "aculeatum"       "adscendens"      "adusta"          "aeruginascens"
##   [9] "agelaea"         "agrimonoides"    "aipolia"         "alba"
##  [13] "albella"         "alpigena"        "alpina"          "amara"
##  [17] "amygdaloides"    "anagyroides"     "aparine"         "applanatum"
##  [21] "aquifolium"      "aquilinum"       "argena"          "argentata"
##  [25] "aria"            "arvense"         "arvensis"        "atra"
##  [29] "aucuparia"       "australis"       "autumnalis"      "avium"
##  [33] "baccata"         "bella-donna"     "betulinus"       "bifolia"
##  [37] "biziana"         "bulbifera"       "bulbiferum"      "campestre"
##  [41] "capillaris"      "caprea"          "carpinea"        "carpinifolia"
##  [45] "carpophila"      "cepaea"          "cerina"          "cerinelloides"
##  [49] "cervinus"        "chamaedrys"      "chlarotera"      "chlorococcum"
##  [53] "ciliaris"        "citrina"         "coccodes"        "collina"
##  [57] "columnae"        "communis"        "compressa"       "corniculatus"
##  [61] "cornucopiae"     "corymbosum"      "cracca"          "crinalis"
##  [65] "cristatus"       "cymbalaria"      "cyparissias"     "damasonium"
##  [69] "deusta"          "didyma"          "digitata"        "dioica"
##  [73] "disciformis"     "distorta"        "divulsa"         "dumeticola"
##  [77] "ebulus"          "echioides"       "effusum"         "elaeochroma"
##  [81] "enneaphyllos"    "erythraea"       "etrusca"         "eupatoria"
##  [85] "europaea"        "europaeus"       "excelsior"       "faginea"
##  [89] "falcatum"        "farinacea"       "filix-mas"       "flacca"
```

```
##  [93] "flavida"         "flavorubescens" "flexuosa"       "fomentarius"
##  [97] "forsteri"        "fragiforme"     "fragilis"       "fraxinea"
## [101] "gibbosa"         "glabra"         "glabratula"     "glomerata"
## [105] "glycyphyllos"    "golaka"         "graeca"         "grandiflora"
## [109] "griseovirens"    "hagenii"        "hederifolia"    "helix"
## [113] "helleborine"     "heptaphylla"    "heterophylla"   "hieracioides"
## [117] "hirsuta"         "hirsutum"       "hirta"          "hirtus"
## [121] "hypoglossum"     "hypophegea"     "hypoxylon"      "idaeus"
## [125] "insignis"        "intumescens"    "italica"        "kitaibelii"
## [129] "laevigata"       "laevipes"       "lanuginosus"    "lappa"
## [133] "lathyroides"     "latifolium"     "laureola"       "leioplaca"
## [137] "leptalea"        "longifolia"     "longipes"       "lucidum"
## [141] "lupulina"        "lutetiana"      "maculata"       "maculatum"
## [145] "marginata"       "martagon"       "menthifolium"   "mezereum"
## [149] "micrantha"       "microphylla"    "minus"          "monogyna"
## [153] "montana"         "montanum"       "moriformis"     "moschatellina"
## [157] "multiflora"      "muralis"        "murorum"        "naegelii"
## [161] "nemoralis"       "nemorensis"     "nepeta"         "nidus-avis"
## [165] "nitida"          "nobilis"        "nodosum"        "nummularia"
## [169] "nummularium"     "ochracea"       "ochroleucon"    "odoratum"
## [173] "officinalis"     "opalus"         "pallescens"     "paniculata"
## [177] "paradoxa"        "parietina"      "pendulina"      "perforatum"
## [181] "persicifolia"    "pertusa"        "pilosa"         "pinicola"
## [185] "platanoides"     "platyphylla"    "podagraria"     "polymorpha"
## [189] "pratense"        "pratensis"      "pseudoplatanus" "purpurea"
## [193] "purpureum"       "pyracea"        "radiata"        "ramosa"
## [197] "rapunculus"      "reichenbachiana" "relicinum"     "remota"
## [201] "renati"          "repandum"       "repens"         "reptans"
## [205] "rigidula"        "robertianum"    "robustius"      "rotundifolia"
## [209] "rotundifolium"   "rubra"          "rupestre"       "ruta-muraria"
## [213] "scoparius"       "secunda"        "serpens"        "sessilifolium"
## [217] "setiferum"       "sp."            "speciosa"       "spicata"
## [221] "spp."            "stellaris"      "stigma"         "stricta"
## [225] "strictus"        "subtomentosum"  "subtomentosus"  "sulcata"
## [229] "sylvatica"       "sylvaticum"     "sylvestris"     "tanfanii"
## [233] "trachelium"      "trichomanes"    "trinervia"      "trivialis"
## [237] "tuberaster"      "uniflora"       "ursinum"        "varius"
## [241] "vernus"          "verrucosum"     "versicolor"     "vesca"
## [245] "villosus"        "virgaurea"      "viridis"        "viscosum"
## [249] "vitalba"         "vulgare"        "vulgaris"
```

```r
#check incerta
incerta_index <- c("", "sp.", "spp.")

incerta <- data |>
  filter(epit %in% incerta_index) |>
  distinct(species, genus, taxon)

incerta
```

```
## # A tibble: 19 x 3
##    species         genus      taxon
##    <chr>           <chr>      <fct>
##  1 Carduus sp.     Carduus    Vascular plants
```

```
##  2 Prunus sp.      Prunus      Vascular plants
##  3 Thesium sp.     Thesium     Vascular plants
##  4 Verbascum sp.   Verbascum   Vascular plants
##  5 Caloplaca sp.   Caloplaca   Epiphytic lichens
##  6 Gyalecta sp.    Gyalecta    Epiphytic lichens
##  7 Lepraria sp.    Lepraria    Epiphytic lichens
##  8 Opegrapha sp.   Opegrapha   Epiphytic lichens
##  9 Parmelina spp.  Parmelina   Epiphytic lichens
## 10 Physcia sp.     Physcia     Epiphytic lichens
## 11 Armillaria sp.  Armillaria  Saproxylic fungi
## 12 Coniophora sp.  Coniophora  Saproxylic fungi
## 13 Crustoderma sp. Crustoderma Saproxylic fungi
## 14 Hypoxylon sp.   Hypoxylon   Saproxylic fungi
## 15 Scutellinia sp. Scutellinia Saproxylic fungi
## 16 Stereum sp.     Stereum     Saproxylic fungi
## 17 Trametes sp.    Trametes    Saproxylic fungi
## 18 Xylaria sp.     Xylaria     Saproxylic fungi
## 19 Xylariaceae sp. Xylariaceae Saproxylic fungi
```

```r
#check congenerics of incerta

data |>
  filter(genus %in% incerta$genus) |>
  distinct(species, taxon) |>
  print(n = 50)
```

```
## # A tibble: 38 x 2
##    species               taxon
##    <chr>                 <fct>
##  1 Carduus sp.           Vascular plants
##  2 Prunus avium          Vascular plants
##  3 Prunus sp.            Vascular plants
##  4 Thesium sp.           Vascular plants
##  5 Verbascum sp.         Vascular plants
##  6 Caloplaca cerina      Epiphytic lichens
##  7 Caloplaca sp.         Epiphytic lichens
##  8 Gyalecta sp.          Epiphytic lichens
##  9 Lepraria rigidula     Epiphytic lichens
## 10 Lepraria sp.          Epiphytic lichens
## 11 Opegrapha sp.         Epiphytic lichens
## 12 Parmelina spp.        Epiphytic lichens
## 13 Physcia adscendens    Epiphytic lichens
## 14 Physcia aipolia       Epiphytic lichens
## 15 Physcia biziana       Epiphytic lichens
## 16 Physcia leptalea      Epiphytic lichens
## 17 Physcia sp.           Epiphytic lichens
## 18 Physcia stellaris     Epiphytic lichens
## 19 Armillaria sp.        Saproxylic fungi
## 20 Coniophora sp.        Saproxylic fungi
## 21 Crustoderma sp.       Saproxylic fungi
## 22 Hypoxylon fragiforme  Saproxylic fungi
## 23 Hypoxylon sp.         Saproxylic fungi
## 24 Scutellinia sp.       Saproxylic fungi
## 25 Stereum hirsutum      Saproxylic fungi
```

```
## 26 Stereum sp.           Saproxylic fungi
## 27 Stereum subtomentosum Saproxylic fungi
## 28 Trametes gibbosa      Saproxylic fungi
## 29 Trametes hirsuta      Saproxylic fungi
## 30 Trametes ochracea     Saproxylic fungi
## 31 Trametes sp.          Saproxylic fungi
## 32 Trametes versicolor   Saproxylic fungi
## 33 Xylaria carpophila    Saproxylic fungi
## 34 Xylaria hypoxylon     Saproxylic fungi
## 35 Xylaria longipes      Saproxylic fungi
## 36 Xylaria polymorpha    Saproxylic fungi
## 37 Xylaria sp.           Saproxylic fungi
## 38 Xylariaceae sp.       Saproxylic fungi
```

```
#those that have no congenerics

keep_genus <- data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n == 1)

keep_genus
```

```
## # A tibble: 11 x 4
##    genus       species         taxon                 n
##    <chr>       <chr>           <fct>             <int>
##  1 Carduus     Carduus sp.     Vascular plants       1
##  2 Thesium     Thesium sp.     Vascular plants       1
##  3 Verbascum   Verbascum sp.   Vascular plants       1
##  4 Gyalecta    Gyalecta sp.    Epiphytic lichens     1
##  5 Opegrapha   Opegrapha sp.   Epiphytic lichens     1
##  6 Parmelina   Parmelina spp.  Epiphytic lichens     1
##  7 Armillaria  Armillaria sp.  Saproxylic fungi      1
##  8 Coniophora  Coniophora sp.  Saproxylic fungi      1
##  9 Crustoderma Crustoderma sp. Saproxylic fungi      1
## 10 Scutellinia Scutellinia sp. Saproxylic fungi      1
## 11 Xylariaceae Xylariaceae sp. Saproxylic fungi      1
```

These should be kept because they are the only ones in their genus. Now, let's check the ones with more entries:

```
data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n > 1) |>
  print(n = 50)
```

```
## # A tibble: 27 x 4
##    genus     species                taxon                 n
##    <chr>     <chr>                  <fct>             <int>
##  1 Prunus    Prunus avium           Vascular plants       2
```

```
##   2 Prunus    Prunus sp.            Vascular plants       2
##   3 Caloplaca Caloplaca cerina      Epiphytic lichens     2
##   4 Caloplaca Caloplaca sp.         Epiphytic lichens     2
##   5 Lepraria  Lepraria rigidula     Epiphytic lichens     2
##   6 Lepraria  Lepraria sp.          Epiphytic lichens     2
##   7 Physcia   Physcia adscendens    Epiphytic lichens     6
##   8 Physcia   Physcia aipolia       Epiphytic lichens     6
##   9 Physcia   Physcia biziana       Epiphytic lichens     6
##  10 Physcia   Physcia leptalea      Epiphytic lichens     6
##  11 Physcia   Physcia sp.           Epiphytic lichens     6
##  12 Physcia   Physcia stellaris     Epiphytic lichens     6
##  13 Hypoxylon Hypoxylon fragiforme  Saproxylic fungi      2
##  14 Hypoxylon Hypoxylon sp.         Saproxylic fungi      2
##  15 Stereum   Stereum hirsutum      Saproxylic fungi      3
##  16 Stereum   Stereum sp.           Saproxylic fungi      3
##  17 Stereum   Stereum subtomentosum Saproxylic fungi      3
##  18 Trametes  Trametes gibbosa      Saproxylic fungi      5
##  19 Trametes  Trametes hirsuta      Saproxylic fungi      5
##  20 Trametes  Trametes ochracea     Saproxylic fungi      5
##  21 Trametes  Trametes sp.          Saproxylic fungi      5
##  22 Trametes  Trametes versicolor   Saproxylic fungi      5
##  23 Xylaria   Xylaria carpophila    Saproxylic fungi      5
##  24 Xylaria   Xylaria hypoxylon     Saproxylic fungi      5
##  25 Xylaria   Xylaria longipes      Saproxylic fungi      5
##  26 Xylaria   Xylaria polymorpha    Saproxylic fungi      5
##  27 Xylaria   Xylaria sp.           Saproxylic fungi      5
```

Of those, *Prunus* is kept because it refers to seedlings which are distinct from those of *Prunus avium*. *Lepraria* is kept because it was different from *Lepraria rigidula*, while *Physcia* and *Caloplaca* should be removed.

```r
keep_genus <- c(keep_genus$genus, "Lepraria", "Prunus")

#keep only those incerta
incerta <- incerta |>
  filter(!(genus %in% keep_genus))

data <- data |>
  anti_join(incerta)
```

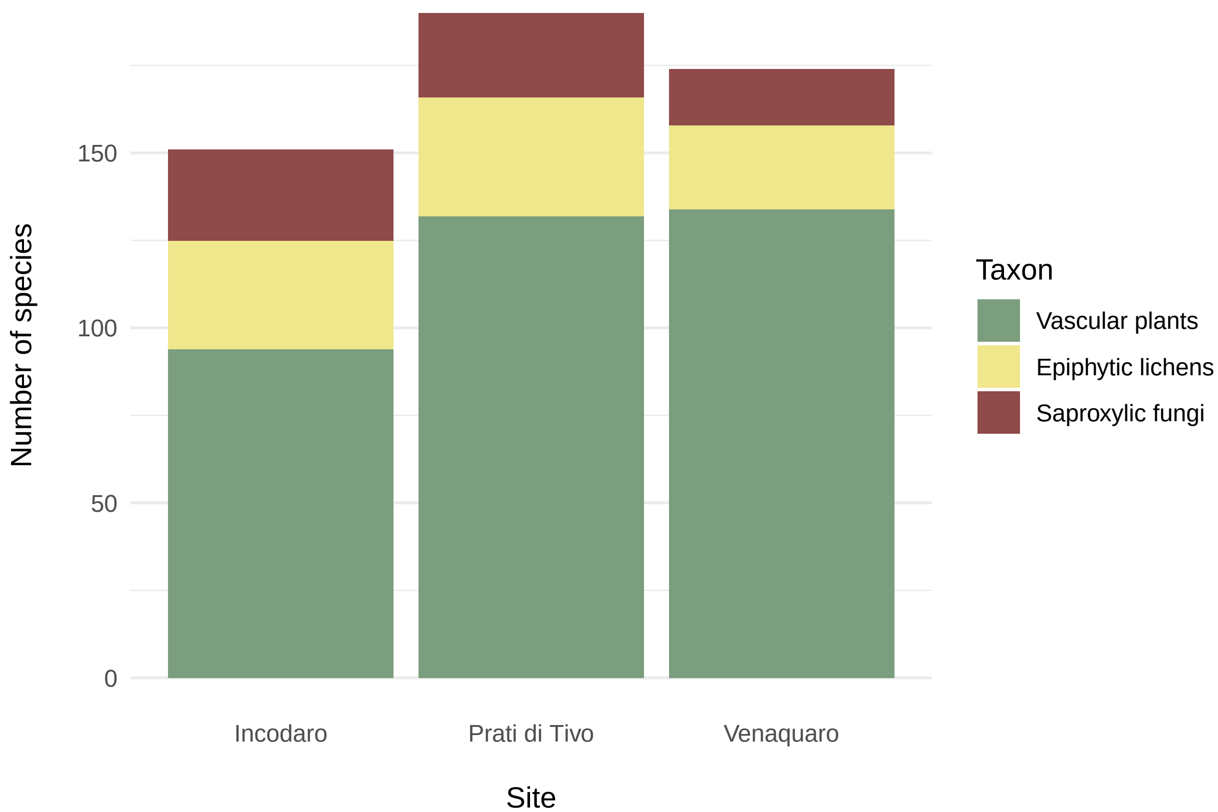
```
## Joining with `by = join_by(taxon, species, genus)`
```

```r
# plot! ------------------------------------------------------------------

richness <- data |>
  group_by(site, taxon) |>
  summarise(rich = n_distinct(species), .groups = "drop")

richness
```

```
## # A tibble: 9 x 3
##   site          taxon              rich
##   <fct>         <fct>             <int>
## 1 Incodaro      Vascular plants      94
```

```
## 2 Incodaro      Epiphytic lichens    31
## 3 Incodaro      Saproxylic fungi     26
## 4 Prati di Tivo Vascular plants     132
## 5 Prati di Tivo Epiphytic lichens    34
## 6 Prati di Tivo Saproxylic fungi     24
## 7 Venaquaro     Vascular plants     134
## 8 Venaquaro     Epiphytic lichens    24
## 9 Venaquaro     Saproxylic fungi     16
```

```r
richness |>
  ggplot(aes(x = site, y = rich, fill = taxon)) +
  geom_col(position = position_stack(reverse = TRUE)) +
  scale_fill_discrete(palette = c("#7B9E7E", "#F0E68C", "#8F4A4A")) +
  theme_minimal() +
  labs(
    x = "Site",
    y = "Number of species",
    fill = "Taxon"
  ) +
  theme(
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15)),
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold", margin = margin(b = 15)),
    panel.grid.major.x = element_blank()
  )
```



```r
ggsave("richness_stack.png", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
ggsave("richness_stack.eps", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
```